



Differentiability : RACE-01

Single Correct Question

1. A function is defined as follows

$$f(x) = \begin{cases} x^3 & ; \quad x^2 < 1 \\ x & ; \quad x^2 \geq 1 \end{cases} \text{ is}$$

- (A) Continuous but not differentiable at $x = 1$
 (B) Discontinuous but differentiable at $x = 1$
 (C) Discontinuous but not differentiable at $x = 1$
 (D) Continuous but differentiable at $x = 1$

MDY132

2. If $f(x) = \begin{cases} \ln x & \text{for } x \geq e^2 \\ ax + b & \text{for } x < e^2 \end{cases}$ is differentiable at $x = e^2$, then find value of a and b .

- (A) $a = \frac{3}{e^2}, b = 1$ (B) $a = \frac{1}{e^2}, b = 2$ (C) $a = \frac{1}{e^2}, b = 1$ (D) $a = \frac{1}{e^2}, b = 5$

MDY133

3. If $f(x) = \begin{cases} x & , \quad x \leq 1 \\ x^2 + bx + c & , \quad x > 1 \end{cases}$ then find the values of b and c if $f(x)$ is differentiable at $x = 1$

- (A) $b = -2, c = 1$ (B) $b = -2, c = -2$ (C) $b = -1, c = 2$ (D) $b = -1, c = 1$

MDY134

4. A function is defined as follows $f(x) = (x-1)|x-1|$.

- (A) Continuous & not differentiable at $x = 1$
 (B) Continuous & differentiable at $x = 1$
 (C) Discontinuous & differentiable at $x = 1$
 (D) Discontinuous & not differentiable at $x = 1$

MDY135

5. Which of the following functions is differentiable at $x = 0$?

- (A) $\cos(|x|) + |x|$ (B) $\cos(|x|) - |x|$ (C) $\sin(|x|) + |x|$ (D) $\sin(|x|) - |x|$

MDY136

6. If $f(x) = x(\sqrt{x} - \sqrt{x+1})$, then indicate the correct alternative(s):

- (A) $f(x)$ is continuous but not differentiable at $x = 0$
 (B) $f(x)$ is differentiable at $x = 0$
 (C) $f(x)$ is not differentiable at $x = 0$
 (D) none

MDY137

Multiple Correct Question

7. If $f(x) = \begin{cases} ax^3 + b, & \text{for } 0 \leq x \leq 1 \\ 2\cos \pi x + \tan^{-1} x, & \text{for } 1 < x \leq 2 \end{cases}$ be the differentiable function at $x = 1$, then find a and

b . (where $[.]$ denotes the greatest integer function)

- (A) $a = \frac{1}{3}$ (B) $b = \frac{\pi}{4} - \frac{13}{6}$ (C) $a = \frac{1}{6}$ (D) $b = \frac{\pi}{4} - \frac{13}{3}$

MDY138

8. Select the correct statements-

- (A) $f(x) = (x+1)^{1/3}$ is not differential at $x = -1$
 (B) $f(x) = 3|x-1|$ is not differential at $x = 1$
 (C) $f(x) = [x+1]$ is not differential at $x = 0$ (where $[.]$ denotes the greatest integer function)
 (D) $f(x) = x^2 + 2x + 1$ is not differential at $x = 0$

MDY139

9. The function $f(x) = \begin{cases} x^3 - 1 & ; \quad 1 < x < \infty \\ x - 1 & ; \quad -\infty < x \leq 1 \end{cases}$ is -

- (A) continuous at $x = 1$
 (B) Differentiable at $x = 1$
 (C) Discontinuous at $x = 1$
 (D) Not differentiable at $x = 1$

MDY140

10. A function is defined as follows

$$f(x) = \begin{cases} 1 & , \quad -\infty < x < 0 \\ 1 + |\sin x| & , \quad 0 \leq x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2 & , \quad \frac{\pi}{2} \leq x < \infty \end{cases}$$

- (A) Non-derivable at $x = 0$ (B) continuous at $x = 0$
 (C) derivable and continuous at $x = \frac{\pi}{2}$ (D) derivable and discontinuous at $x = \frac{\pi}{2}$

MDY141

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	D	B	D	B	B,C	A,B,C	A,D	A,B,C



Differentiability : RACE-02

Single Correct Question

1. A function $f: \mathbb{R} \rightarrow (0, \infty)$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$. If $f(x) \neq 0$ for any x , is differentiable on $\mathbb{R}$ and $f'(0) = 2$ then find $f(x)$

(A) e^{-x} (B) e^{2x} (C) e^x (D) e^{-2x}

MDY142

2. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the condition $f\left(\frac{x+y}{k}\right) = \frac{f(x)+f(y)}{k}$ ($k \neq 0, 2$). If $f(x)$ is differentiable on $\mathbb{R}$ and $f'(0) = m$ then find $f(x)$

(A) mx (B) $x^2 + mx$ (C) $mx + 1$ (D) $x^3 + mx$

MDY143

Multiple Correct Question

3. A function is defined as follows $f(x) = \begin{cases} [x] & ; 0 < x \leq 2 \\ 2x-2 & ; 2 < x < 3 \end{cases}$, where $[.]$ denotes greatest integer function.

(A) continuous and not differential at $x = 1$
 (B) discontinuous and not differential at $x = 2$
 (C) discontinuous and not differential at $x = 1$
 (D) discontinuous and differential at $x = 2$

MDY144

4. The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & |x| > 1 \end{cases}$. Discuss the continuity and differentiability of $f(x)$:

(A) continuous on $\mathbb{R} - \{1, -1\}$ (B) continuous on $\mathbb{R} - \{1\}$
 (C) Differentiable on $\mathbb{R} - \{-1, 1\}$. (D) Differentiable on $\mathbb{R} - \{1\}$.

MDY145

5. Suppose f is a derivable function that satisfies the equation $f(x+y) = f(x) + f(y) + x^2y + xy^2$ for all real numbers x and y . Where $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$ then find

(A) $f(0) = 0$ (B) $f'(0) = 0$ (C) $f'(x) = x^2 + 1$ (D) $f'(1) = 2$

MDY146

Integer Type Question

6. Let $f(x) = \min \{|x-1|, |x+1|, 1\}$. Find the number of points where it is not differentiable. **MDY147**
7. Consider function $f(x) = \min \{\tan x, \cot x\}$ in $(0, 2\pi)$. Number of points where f is either fails to be derivable is ' m ' and number of points where it is discontinuous is ' n '. Then the value of $m - n$ is **MDY148**
8. Let $f(x) = \max \{2\sin x, 1 - \cos x\} \forall x \in (0, \pi)$. If $f(x)$ not differentiable at $x = \pi - \cos^{-1} \frac{k}{5}$ then the value of k is **MDY149**
9. $f(x) = \begin{cases} |1-4x^2| & , 0 \leq x < 1 \\ [x^2-2x] & , 1 \leq x < 2 \end{cases}$ then number of points where $f(x)$ is non-derivable in $x \in (0, 2)$ is equal to **MDY150**
10. $f(x) = |[x] x|$ then number of points where $f(x)$ is non-derivable in $x \in [-1, 2]$ is equal to (where $[.]$ denotes greatest integer function) **MDY151**

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B,C	B,C	A,B,C,D	5	4	3	2	3